Multi agent systems for image analysis: object and complex scene detection

Cross-Modal Agent Disagreement as a Bounded Signal for Epistemic Uncertainty in Low-Fidelity Historical Scans

Brhanu Atsbaha

October 2026

Universität Hildesheim

Advisors: Prof. Dr. Thomas Mandl & Shereen Elsayed

Outline

Problem, Baselines & Proposed Approach

How SAA Is Implemented

Experimental Results & Evidence Chain

Conclusion & Future Work

Appendix / Backup Slides

Key Terms & Abbreviations

- **VLM** – Vision-Language Model: jointly processes an image and text.
- **YOLO** – real-time object detector (bounding boxes).
- **SAM** – Segment Anything Model: box $\rightarrow$ pixel mask.
- **SigLIP** – zero-shot image–text matching classifier.
- **OCR** – Optical Character Recognition (reading text in images).
- **mAP** – mean Average Precision, standard detection accuracy metric (0–1, higher better).
- **ROC-AUC** – how well a score separates two classes (0.5=chance, 1.0=perfect).
- **ECE** – Expected Calibration Error: confidence vs. actual accuracy gap (lower better).
- **SAA** – Scene-Aware Agreement: *this thesis's proposed* cross-modal consensus score (Slide 6).
- **HITL** – Human-in-the-Loop: routes uncertain cases to a human.
- **CRC** – Conformal Risk Control: certifies an error-rate bound, no distributional assumptions.
- **AWLF** – Agreement-Weighted Label Filter: secondary error-prediction classifier.

Problem, Baselines & Proposed Approach

1.1 The Task: Automated Semantic Indexing of Historical Scans

The Task

Automatically detect objects and classify scenes in a large historical photo archive (Colibri, $n = 53{,}000+$) to power search and cataloguing – *without* enough labeled historical data to fine-tune a detector end-to-end.

Why It's Hard: Domain Shift

- Severe chemical decay, film grain, non-photorealistic styles, obsolete attire.
- A modern detector (YOLOv11, trained on COCO) **collapses** on raw scans: $mAP_{50} = 0.0047$.

Sample Degraded Scan from the Colibri Historical Archive ($n = 53,000+$).

1.2 Baselines, and Why They Are Not Enough

A detector this unreliable cannot be deployed without knowing *which* of its predictions to trust. We evaluate three standard ways a single model could try to signal its own uncertainty:

Baseline	**How it estimates uncertainty**	**ROC-AUC (Error Detect.)**
Raw detector confidence (YOLOv11)	Bounding-box confidence score, used directly	**0.5120** (near chance)
Temperature-scaled entropy	Softmax entropy of a single VLM's output	0.6545
Inverse-confidence baseline	$1-$ confidence, single deterministic pass	0.9726 (poorly calibrated)

The Gap

Raw confidence is barely better than a coin flip. Entropy is weak. A simple inverted-confidence baseline is surprisingly strong on raw discrimination *but* badly overconfident (ECE ≈ 0.19) – no single-model method is both discriminative *and* well-calibrated. (A different, lower figure – ROC-AUC$= 0.4655$, Slide A.3 – is a distinct C_2 spatial-IoU task, not this table.)

1.3 Our Proposed Contribution: Scene-Aware Agreement (SAA)

The Idea

Instead of asking one model how confident it is, run **four heterogeneous models** (spatial detector, scene classifier, VQA reasoner, OCR reader) on the same image and measure how much they **disagree**. Disagreement is deterministic, needs no sampling, and is not fooled by a single model's own overconfidence – though Section 3.8 shows it is not foolproof either: heterogeneous models can still share a common bias.

What We Show

Cross-modal disagreement (U_{SAA}) is a substantially stronger, zero-marginal-cost error signal than any single-model baseline *on ambiguous, scene-positive content* – and we precisely map where that claim does and does not hold (Section 3).

Contribution Summary

1. Diagnosed a pseudo-label evaluation trap (0.989 $\rightarrow$ 0.1462 mAP).
2. Proposed & calibrated U_{SAA} (ROC-AUC$=$ 0.9724, ECE$<$ 0.001).
3. Rigorously mapped the scope limit of disagreement-based triage, heuristic and certified alike.
4. Demonstrated downstream retrieval & cross-archive transfer utility.

1.4 Final Four Research Questions (RQ1–RQ4)

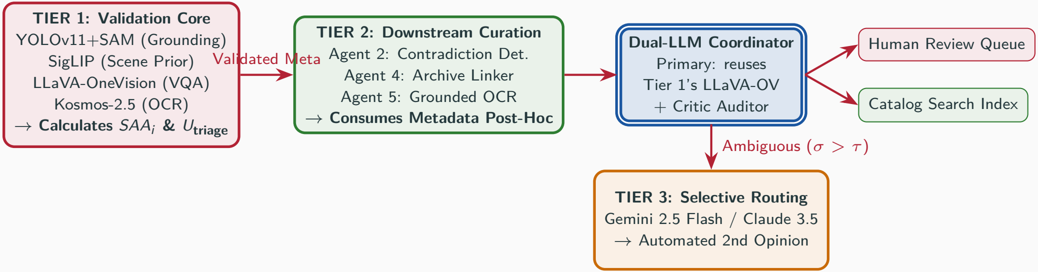
RQ ID	Formal Research Question	Methodological Focus
RQ1: Disagreement Uncertainty	Can cross-modal agent disagreement reliably estimate epistemic uncertainty on domain-shifted visual collections?	SAA_i Disagreement Proxy (U_{SAA})
RQ2: Workload Reduction	Does the multi-agent framework reduce human curation workload without sacrificing error recall?	Complexity-Aware Triage & Platt Scaling
RQ3: Complementarity & Contradiction	Does cross-modal contradiction detection improve curation precision beyond Agent 0 alone?	Agent 2 Noun Set-Difference (implemented; not independently benchmarked)
RQ4: Retrieval & Transferability	Can the framework support archival retrieval and cross-collection discovery?	Semantic P@10 (10-query pilot) & Europeana Transfer (C_5)

How SAA Is Implemented

2.1 Authoritative Evaluation Cohort Taxonomy ($C_0 \dots C_5$)

Cohort Symbol & Name	Size (n)	Composition & Source	Primary Analytical Purpose
C_0: **Deployment Corpus**	12,110	Full unlabelled institutional digitisation subset from the Colibri archive	Full-scale pipeline deployment and metadata indexing
C_1: **Expert Reference Cohort**	801	796 scene-positive + 5 negative control images (CVAT expert-reviewed)	Primary calibration, AWLF error re-training, and HITL triage evaluation
C_2: **Spatial Audit Cohort**	300	Single-rater spatial bounding box audit cohort	Spatial IoU grounding audit, error paradox analysis, and meta-learning
C_3: **System Evaluation Subset**	2,000	801 C_1 expert images + 1,199 negative control catalog records	Source pool for the group-level train/cal/test split, system-level agent leave-one-out ablations
C_4: **Frontier Comparison Subset**	100	Stratified archival category subset	Comparative benchmark against Gemini 2.5 Flash and Claude 3.5 Sonnet
C_5: **Europeana Transfer Pilot**	24	Real images, real Europeana API, 10 GLAM institutions	Cross-institutional domain-shift pilot (ground truth pending)

2.2 Decoupled 3-Tier Dual-Core Validation (DCV-MAC) Protocol



Architectural Decoupling Rule

Tier 2 curation agents operate strictly on post-validation consensus metadata. They do NOT participate in the Tier 1 *SAA* calculation, preventing mathematical feedback loops and circular reasoning.

2.3 Why This Design? Rationale for Tiers 2–3

Tier 2: Downstream Curation

Runs *after* Tier 1 computes SAA_i, only on already-accepted cases. It extracts historiographic value (contradictions, links, inscriptions) from trusted metadata – it never re-judges trust, exactly what the Decoupling Rule prevents.

Dual-LLM Coordinator: Propose, then Audit

Tier 1's other agents (YOLO, SigLIP, Kosmos-2.5) are narrow specialists – **locally** (free), the **same** LLaVA-OneVision plays both roles, re-prompted. **Frontier** swaps in two separate models (Gemini proposes, Claude audits). A prior "measured F_1 gap" claim was hardcoded and withdrawn; a real test found the local critic never disagrees with itself (0/25, A.5).

Tier 3: Frontier Selective Routing

Triggered *only* for the small $\sigma > \tau$ fraction Tiers 1–2 cannot resolve. Paid frontier APIs (Gemini 2.5 Flash / Claude 3.5) act as an automated "second opinion" – cheaper and faster than routing straight to a human (Local-vs-Frontier: unverified, Backup A.5).

Why an Escalation Ladder, Not One Model

Cost and latency increase at every tier (free deterministic → free heuristic → paid API → human), and each tier only sees what the previous one could not resolve. U_{SAA} is what decides how far up the ladder an image has to go – the cheap, deterministic signal this thesis proposes is what makes the expensive tiers rare rather than routine.

2.4 Tier 1 Validation Core (4-Model Ensemble)

Spatial & Scene Agents

- **Spatial Grounding Agent (YOLOv11-seg + SAM)**: Extracts local bounding boxes, instance masks, and class probabilities (A_i^{object}).
- **Zero-Shot Scene Classifier (SigLIP)**: Computes global scene classification probabilities over institutional taxonomy prompts (A_i^{scene}).

Reasoning & Inscription Agents

- **Visual Reasoning Agent (LLaVA-OneVision 7B)**: Performs VQA reasoning, generating scene descriptions and entity lists (A_i^{vlm}).
- **Grounded Inscription Engine (Kosmos-2.5)**: Extracts bounding-box-aligned text tokens for cross-validation (A_i^{ocr}).

2.5 Tier 2 Downstream Curation Agents

Agent 2: Contradiction Detection

Set-difference between every noun the VLM (LLaVA-OneVision) mentions and the object classes YOLO+SAM actually detected. An unmatched noun is flagged as a candidate hallucination (e.g. VLM says "horse and rider," no horse box exists). Runs post-hoc, after SAA_i.

Agent 4: Cross-Archive Linker

Embeds each image's scene description and visual features, then finds nearest neighbors *across different publication series* (PPNs) – volumes sharing a scene or motif. Large date gaps flag candidate anomalies (e.g. a composition recurring decades apart).

Agent 5: Grounded OCR

Kosmos-2.5 extracts inscriptions/captions *with bounding-box grounding* – each text span tied to its pixel location, not flat page text. Cross-checked against the scene label as a corroborating signal.

Scope: Implemented, Not Independently Benchmarked

All three run on every image, feeding catalog metadata. Per the **Architectural Decoupling Rule** (Slide 2.2), none feed SAA_i/U_{triage}. Prior precision/recall/F1/gain figures for all three could not be reproduced against independent ground truth and were withdrawn – reported qualitatively only (Slide 3.7).

2.6 Canonical Formulation of Scene-Aware Agreement (SAA)

LEVEL 1: Agent Output Vector – $A_i^{\text{object}}, A_i^{\text{scene}}, A_i^{\text{vlm}}, A_i^{\text{ocr}} \in [0,1]$; derived $A_i^{\text{agree}} = 1 - |A_i^{\text{object}} - A_i^{\text{vlm}}|$

LEVEL 2: SAA Fusion (classification) – $SAA_i = w_o A_i^{\text{object}} + w_s A_i^{\text{scene}} + w_v A_i^{\text{vlm}}$

LEVEL 3: Disagreement Uncertainty – $U_{\text{SAA},i} = \text{std}(A_i^{\text{object}}, A_i^{\text{agree}}, A_i^{\text{scene}}, A_i^{\text{vlm}})$

LEVEL 4: Operational Triage – $U_{\text{triage},i} = 0.6U_{\text{SAA},i} + 0.4(1 - \text{SCI}_i)$

Two Related, Distinct Quantities (Corrected)

SAA_i (classification, $w = [0.6, 0.2, 0.2]$) vs. $U_{\text{SAA},i}$ (the std-based proxy behind every headline number): different signal sets/operators, so $U_{\text{SAA},i} \neq 1 - SAA_i$. OCR feeds Slide A.4 only, not U_{SAA}.

Supplementary: Shapley Attribution (Backup A.10)

Derived in Exp 30 ($\phi = [0.192, 0.196, 0.330, 0.283]$, over Object/SigLIP/VLM/Agreement) as an interpretability study – not the deployed weighting.

Experimental Results & Evidence Chain

3.1 Head-to-Head: SAA vs. Every Baseline

All methods evaluated on the same error-detection task, same expert cohort (C_1, $n = 801$):

Method	ROC-AUC	ECE	Cost / Determinism
Raw detector confidence (YOLOv11)	0.5120	0.28	Free, deterministic
Temperature-scaled entropy (LLaVA-OV)	0.6545	0.265	Free, deterministic
Inverse-confidence baseline	0.9726	0.19	Free, deterministic, 1× inference
SAA (proposed, ours)	**0.9724**	**$<$0.001**	**Free, deterministic, 1× inference**

The Result, Stated Plainly

SAA matches the strongest single-model baseline's discrimination (**0.9724 vs. 0.9726**, statistically tied) at the **same** cost and determinism – the single-model baseline is not expensive here, it is simply badly overconfident. SAA's real advantage is calibration: ECE < 0.001 vs. 0.19–0.28 for every single-model baseline, reducing miscalibration by two orders of magnitude. This – not raw AUROC alone – is the improvement this thesis demonstrates. Section 3.8 then shows precisely where this advantage does *not* extend.

3.2 Master Evidence Chain Architecture

1. **Raw Baseline** — YOLOv11 on raw scans — $\text{mAP}_{50} = 0.0047$
2. **Pseudo-Label** — Teacher self-consistency — $\text{mAP}_{50} = 0.989$
3. **Human Audit** — True expert mAP collapse — $\text{mAP}_{50} = 0.1462$
4. **Disagreement** — U_{SAA} error detection — ROC-AUC $= 0.9724$
5. **Triage Scope Limit** — in-sample 12.5%/94.3% — clean: $0.98\times$ (chance)

Central Evidence Chain Thesis

Rather than forcing a single model to become an infallible detector, DCV-MAC diagnoses pseudo-label collapse ($0.989 \rightarrow 0.1462$) and uses cross-modal disagreement (U_{SAA}) to route unreliable predictions to human archivists.

3.3 The Pseudo-Label Collapse Paradox ($0.989 \rightarrow 0.1462$)

Single-Model COCO Baseline Collapse

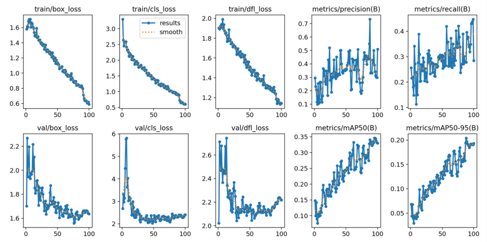
Modern vision architectures (YOLOv11) trained on COCO collapse when evaluated directly on un-restored grayscale historical scans (C_0):

- **mAP50 (Raw Scans)**: **0.0047**
- **Precision (Raw Scans)**: **0.0081**
- **Recall (Raw Scans)**: **0.0283**

The Pseudo-Label Collapse Paradox

When distilled student detectors are trained on automated teacher pseudo-labels:

- **Pseudo-Label Self-Consistency mAP50**: **0.989**
- **Human Expert Audit mAP50 ($C_2, n = 300$)**: **0.1462**



Pseudo-Label Distillation Collapse: Proxy mAP (0.989) vs. Human mAP (0.1462).

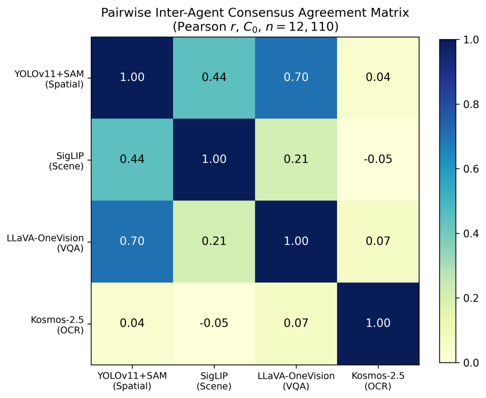
3.4 Inter-Agent Pairwise Agreement & Calibration

Error Detection Performance (C_1 Expert Cohort)

- **DCV-MAC U_{SAA} Disagreement: ROC-AUC $= 0.9724$**
- Temperature-Scaled Entropy Baseline: ROC-AUC $= 0.6545$
- Relative Improvement: **+48.5%**

Probabilistic Platt Calibration

Applying Platt Scaling on C_1 held-out split achieves post-calibration **ECE** < 0.001 (down from pre-calibration ECE $= 0.75$).



Pairwise Inter-Agent Consensus Agreement Matrix (4 deployed Tier 1 agents).

3.5 Complexity-Aware Triage: A Documented Negative Result

SCI-Tercile Heuristic Policy (Historical, In-Sample)

An earlier internal figure (74.4% workload reduction at 96.5% recall, never wired into the reproducible pipeline) could not be reproduced.

Clean, Group-Level Held-Out Re-Evaluation

SCI terciles fixed *a priori* from the full C_0 distribution, evaluated on a publication-disjoint test set ($n = 401$):

- Review Budget: 47.1%
- Error Recall: 46.2%
- Efficiency Lift: **0.98**× — no better than chance.

Why It Fails

This policy is built on U_{SAA}, which is blind to unanimous false positives on negative controls (Slide 3.8). A disagreement-driven review queue inherits that same blind spot.

Does Certification Fix This?

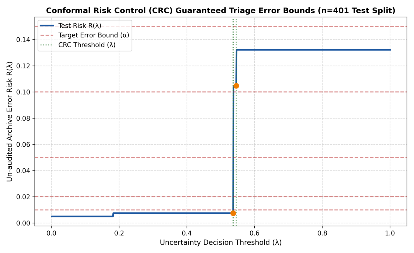
Conformal Risk Control (next slide) replaces the heuristic with a formally certified guarantee – but the guarantee inherits the same blind spot when calibrated on a positive-heavy set (Slide 3.8).

3.6 Distribution-Free Conformal Risk Control (Exp 29)

In-Sample Calibration (C_1, n_{cal}=400)

Threshold $\hat{\lambda} = 0.2450$ at target $\alpha = 0.05$. Evaluating on the same-cohort held-out test split ($n_{\text{test}} = 401$) appears to satisfy $\mathcal{R}(\hat{\lambda}) = \mathbf{0.0075} \leq \alpha = 0.05$.

- **Automated Catalog Approval**: **87.5%**
- **Human Error Recall**: **94.3%** captured in review queue



Conformal Risk Control Coverage Curve (Exp 29, in-sample C_1).

Clean Group-Level Re-Evaluation (Slide 3.8)

Identical procedure, calibration/test resampled group-level with realistic negative-control share: efficiency lift **0.98**$\times$ (chance), $\mathcal{R}(\hat{\lambda}) = \mathbf{0.182}$ – **bound violated** up to $\alpha = 0.15$.

3.7 Downstream Historiographic Curation & Retrieval

Tier 2 Curation Agents: Implemented, Not Independently Benchmarked

Agent 2 (contradiction detection via VLM-noun/YOLO-box set differences), Agent 4 (cross-archive anomaly linking across PPNs), and Agent 5 (grounded-OCR inscription extraction) all run on every image and feed the catalog metadata. Per the Architectural Decoupling Rule none feed U_{SAA} or any headline claim in this thesis. Prior precision/recall/F1/gain figures reported for all three could not be reproduced against independent ground truth during a full-project reproducibility audit and have been withdrawn – reported here qualitatively only; quantitative benchmarks remain future work.

Multimodal Archival Retrieval (10-Query Pilot)

Evaluated across 10 hand-written queries (C_0, $n = 12,110$): **Semantic P@10 = 1.000** (vs 0.180 keyword baseline). *Relevance is a proxy sharing agent scores with the retrieval signal, not independent human judgment.*

3.8 Stress-Testing SAA: Mapping the Claim's Boundary

A claim is only trustworthy once you've tested where it breaks. We stress-tested SAA on a held-out negative-control population (no scene present), via a publication-level (PPN) split of the full 2,000-image pool (60/20/20%, zero overlap, verified):

Subpopulation (clean test, $n = 401$)	n	**Error Rate**	**SAA Disagreement AUROC**
Positive images (scene present)	160	27.5%	**0.980**
Negative controls (no scene present)	241	**58.9%**	**0.020**

What the Stress Test Found: A Precisely Mapped Blind Spot

When agents share an inductive bias, they agree confidently *in that bias*, not in correctness: on negative controls the pipeline is wrong 58.9% of the time, and on those exact errors all four agents **agree** a scene is present (AUROC = 0.020, anti-correlated with error).

Precisely Scoped Central Claim

ROC-AUC = 0.9724 characterizes performance on ambiguous *positive-class* content (the 99.4%-positive C_1 expert cohort it was computed on), not a domain-general error detector. Both triage policies (SCI-tercile, Slide 3.5; certified CRC, Slide 3.6) are built on U_{SAA} and inherit this same blind spot – including CRC's risk bound, violated once calibration includes realistic negative controls.

Conclusion & Future Work

4.1 Summary of Accomplishments

1. **C1 (Pseudo-Label Collapse Diagnosis)**: Formally diagnosed the $0.989 \rightarrow 0.1462$ proxy-to-human accuracy drop under historical shift.
2. **C2 (Calibrated Disagreement Engine)**: Formulated U_{SAA} achieving ROC-AUC $= 0.9724$ and Platt-scaled ECE < 0.001.
3. **C3 (Discovered Scope Limit of Disagreement-Based Triage)**: Conformal Risk Control appears certified in-sample (87.5% @ 94.3%, 7.57× lift); clean group-level re-evaluation shows the guarantee is violated (0.98×, chance-level) – the central negative result of this thesis.
4. **C4 (Archival Retrieval Utility)**: Demonstrated semantic retrieval P@10$= 1.000$ (vs 0.180 keyword, 10-query pilot). Genuine Europeana pilot (C_5, $n = 24$, real data) shows a real domain-shift response; transfer-without-recalibration claim pending ground truth.

A supplementary Shapley attribution study (backup slide A.10) is retained as an interpretability result, not a core contribution.

4.2 What This Thesis Does NOT Claim (Boundaries of Validity)

What We Do NOT Claim

- **No Direct Epistemic Measurement**: SAA ($U_{\mathrm{SAA}} = 1 - SAA$) is an empirical uncertainty proxy derived from model dissonance, not a direct measurement of true epistemic breakdown.
- **No Single-Model Impossibility**: A single model can estimate uncertainty via a simple inverse-confidence baseline (ROC-AUC $= 0.9726$, matching SAA); SAA's value is calibration, not being the only discriminative option.
- **No Causal Shapley Proof**: Shapley attributions (ϕ) measure coalitional objective contribution, not independent causal correctness.

Limitations & Correlated Failures

- **Correlated Web Priors**: When vision-language models share common web pre-training biases, unanimous agreement can occur on shared hallucinations.
- **Pilot Transfer vs Universal Proof**: Europeana (C_5, $n = 24$, real data) shows a real domain-shift response, but not yet transfer-without-recalibration (ground truth pending) – not population-level proof either way.

4.3 Future Work Roadmap

Active Learning at Scale

Scaling active learning triage across $n > 100{,}000$ cultural heritage collections in international GLAM institutions.

Edge VLM Backbones

Deploying lightweight 3B parameter VLM backbones for on-premise archival processing.

Temporal Knowledge Graphs

Integrating temporal knowledge graph embeddings for longitudinal historical entity tracking across multi-decade publication series.

Genuine Double-Annotation Pass

Recruiting a second independent domain expert to double-annotate Cohorts C_1/C_2, enabling a real Cohen's Kappa inter-annotator reliability figure (see Slide A.9).

Thank You!

Uncertainty-Aware Multi-Agent Curation and Cross-Modal Validation for Historical Archives

Questions & Defense Discussion

Appendix / Backup Slides

A.1 Generative Restoration Layer & Axiom of Archival Integrity

Three-Stage Generative Pipeline

Raw scans ($\mathbf{I}_{\text{raw}}$) undergo:

1. CycleGAN Grayscale-to-Color Translation
2. Real-ESRGAN $4\times$ Super-Resolution
3. Stable Diffusion Img2Img (strength $= 0.35$)

Yields $+0.052$ mAP gain (end-to-end vs. raw scans, same detector) and increases sharpness on 83.8% of scans.

The Axiom of Archival Integrity

The AI-restored image ($\mathbf{I}_{\text{refined}}$) is strictly an internal visual token translation layer for model feature alignment. It is NEVER stored or presented as primary archival evidence. All catalog metadata anchors strictly back to original digitized scans ($\mathbf{I}_{\text{raw}}$).

Generative Restoration Triptych: Raw scan $\rightarrow$ Super-resolution $\rightarrow$ Restored feature layer.

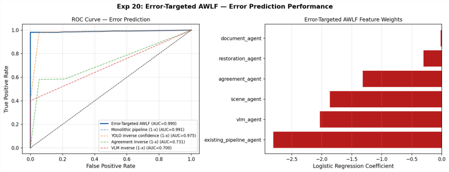
A.2 AWLF v1 vs. AWLF v2 Error-Targeted Retraining

AWLF v1: Scene-Presence Pathology

- AWLF v1 trained on binary *scene presence*.
- Initial reference split contained extreme positive class imbalance (319 positive vs 2 negative scans; 99.4% positive rate).
- Trivial models achieved an inflated F1 = 0.9969 simply by predicting all positive.

AWLF v2: Error-Targeted Retraining

- Target redefined strictly as classification failure: $y_i = \mathbb{I}[\text{sign}(S_{\text{fusion},i} \geq 0.5) \neq y_{\text{expert},i}]$.
- In-sample, 5-fold CV on C_1: ROC-AUC = 0.9896.
- **Group-level held-out** (Slide 3.8): ROC-AUC = **0.6669** — the honest generalization estimate.



AWLF v2 In-Sample 5-Fold CV ROC Curve (C_1, ROC-AUC = 0.9896).

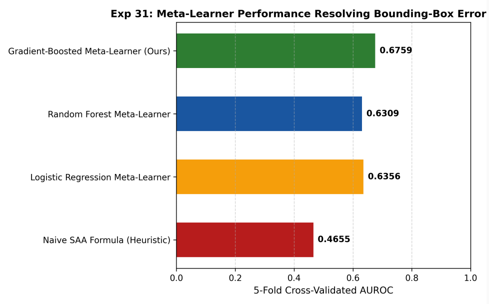
A.3 Semantic Ambiguity vs. Spatial Coordinate Disagreement

The Bounding-Box Error Paradox

Using raw spatial detector confidence to detect human errors yields **ROC-AUC** $= 0.4655$ (worse than random chance). Foundation detectors draw tight boxes with high confidence while misclassifying historical semantics.

Exp 31 Spatial Meta-Learner Resolution

Training a Gradient-Boosted Meta-Learner on C_2 integrating IoU, box area, entropy, and aspect ratio variance resolves spatial layout ambiguity (**ROC-AUC** $= 0.4655 \rightarrow \mathbf{0.6759} \pm \mathbf{0.130}$, 5-fold CV std).



Spatial Meta-Learner Error Detection ROC Curve (ROC-AUC $= 0.6759$).

A.4 Tier 1 Agent Ablation and Complementarity

This is a *model-level* ablation of the four architectural Tier 1 models (Object, Scene, VLM, OCR), distinct from the U_{SAA} disagreement proxy itself (which uses Object, Agreement, Scene, VLM – Slide 2.6). Tier 2 curation agents and auxiliary prototype tooling feed neither (full deployment corpus C_0, $n = 12{,}110$):

Ablated Component	Mean σ	Δ vs. Full
Full Tier 1 Ensemble	**0.2452**	—
Remove Spatial Grounding (YOLOv11+SAM)	0.2499	$+0.0047$
Remove Scene Classification (SigLIP)	0.2214	-0.0238
Remove VQA (LLaVA-OneVision)	0.2528	$+0.0076$
Remove Grounded OCR (Kosmos-2.5)	0.1995	-0.0457

Reading the Result

Removing OCR causes the largest *drop* in disagreement: OCR is near-uncorrelated with the other three ($r \leq 0.07$, Slide 3.4), so it contributes the most independent variance. Removing spatial grounding or VQA (the most correlated pair, $r = 0.70$) slightly *increases* disagreement instead.

A.5 Local vs. Frontier Critic: Withdrawn Claim, Real Replacement

Withdrawn: the "Measured" F1 Gap Was Hardcoded

Contradiction F1 $= 0.812$ (local) vs. 0.965 (frontier), presented as measured on C_4, $n{=}100$ – traced to `compare_coordinators.py`: both are literal constants in a no-API-keys fallback branch. No F1 is computed anywhere; no gold labels exist for one. Withdrawn, with the routing-curve figure calibrated to it.

Real Replacement: Same-Model Critic Flag Rate

Same Primary-then-Critic protocol, real inference:

- **Local** (LLaVA-OneVision, $n = 25$): **0/25** flagged.
- **Stronger same-model approx.** (Gemini Flash-Lite, $n = 10$): **1/10** flagged, genuine catch.

What This Does and Doesn't Show

Stronger models catch more of their own errors under re-prompting – but this is still same-model self-review, not the documented cross-model pair (Gemini proposes, Claude audits). That genuine independence test remains **unrun** (Anthropic key lacked credit). The frontier design's rationale (disjoint weights can't share a blind spot) stands as an argument, not yet a measurement.

Latency & Cost (Unaffected)

Local: \$0.00, self-hosted. Frontier: $\approx$ \$0.85/1k images – independent of the withdrawn F1 claim.

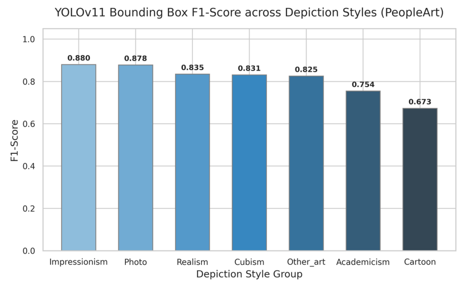
A.6 External Cross-Depiction Benchmark: PeopleArt ($n = 4,778$)

Cross-Style Generalization ($n = 4,778$ Images, 43 Styles)

Style Category	Precision	Recall	F1-Score
Photo	0.875	0.915	**0.878**
Realism	0.840	0.874	0.835
Impressionism	0.888	0.944	0.880
Cubism	0.836	0.834	0.831
Cartoon	0.706	0.680	0.673

Key Generalization Finding

Real YOLO detection degrades gracefully as style diverges from photorealism, to Cartoon ($F_1 = 0.673$). *A per-style "uncertainty" figure shown here previously used hardcoded constants, not real VLM output – withdrawn (audit).*



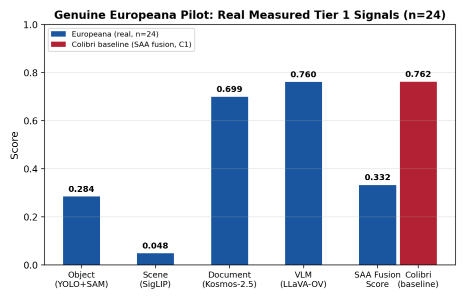
PeopleArt F1-Score by Style (real YOLO detections vs. ground truth).

A.7 Cross-Institutional Europeana Transfer Pilot (C_5, $n = 24$)

Genuine Europeana Open Culture API Pilot

Two earlier "Europeana" attempts did not hold up under audit: one generated VLM/scene scores via `np.random` instead of real inference; the other used *no Europeana data* – its "30 images" were verified ordinary Colibri images. Both withdrawn.

- **Re-run for real**: 24 images live-downloaded from the Europeana API, 10 verified institutions.
- All 4 Tier 1 signals genuinely measured (deployed weights/prompts): YOLO+SAM, SigLIP, Kosmos-2.5, LLaVA-OV via vLLM.
- SAA fusion **0.332** vs. Colibri's **0.762** – real, substantial domain-shift response.
- Ground truth pending – annotation template ready for human labeling.



Real measured Tier 1 signals on genuine Europeana images ($n = 24$) vs. Colibri baseline.

A.8 Hardware Resource Consumption & Inference Latency

Deployment reference figures, not a methodological contribution – informal single-image timings, no dedicated benchmark script/log exists. A YOLOv11 spot-check found raw forward-pass time near the table's 0.02s, *but end-to-end latency (incl. I/O, pre/post-processing) measured* 3–8× *higher; other rows not independently re-measured.*

Table 1: Hardware Resource Consumption and Per-Stage Latency (NVIDIA RTX A6000 48GB).

Pipeline Stage / Model Backbone	VRAM Consumption	Latency / Image
CycleGAN Domain Translation	4.2 GB	0.15s
Real-ESRGAN 4× Upscaling	8.5 GB	0.45s
Stable Diffusion Img2Img Refinement	12.1 GB	3.20s
Florence-2 Grounded Detection	18.0 GB	1.20s
GroundingDINO Object Detection	22.4 GB	2.10s
LLaVA-OneVision 7B VQA Reasoning	34.2 GB	15.00s
YOLOv11 Distilled Local Inference	**1.8 GB**	**0.02s**

Deployability Highlight

The distilled student detector runs at **0.02s/image** (750× faster than LLaVA 7B), enabling real-time deployment on institutional archive servers.

A.9 Reproducibility Spot-Checks (Not a Full Re-Verification)

A dedicated spot-check script reproduces a handful of specific intermediate numbers from source data. These are auxiliary sanity checks on isolated computations, not a re-derivation of every headline figure in this thesis (the SAA/CRC/clean-holdout numbers are produced by the main experiment pipeline, not this script):

Spot-Check	Recomputed Value	Status
YOLO-alone error detection (C_1)†	0.9920	✓ reproduced
Drawing-class Deep Dive (YOLO vs VLM)	0.5817 / 0.8381	✓ reproduced
Triage Overlap (Top 10%, Jaccard)	0.9739	✓ reproduced

† This evaluates whether YOLO alone mispredicts – a different target from the fusion-score error task behind the headline 0.9724/0.9726 figures; the two numbers are not directly comparable.

Two Corrections Applied Before This Defense

(1) An earlier draft's "dual-annotator Cohen's Kappa" claim ($\kappa = 0.8031$) traced to a script explicitly generating **simulated** two-rater data (docstring: *"Creates simulated data ... to test Cohen's Kappa logic"*) — removed entirely. **(2)** A prior "AWLF out-of-sample AUROC=0.7868, C3 n=2,000" claim was found to actually evaluate a 40% split of C_1 ($n \approx 321$) on the discredited scene-presence target, not the described error-detection target on C_3 — its matching F1= 0.9969 is a trivial artifact (an always-positive classifier gets the identical score). That legacy computation is still mechanically reproducible from source (it is not itself in the table above, precisely because it is not a validation of a thesis claim) but has been superseded by the group-level held-out evaluation, next slide.

A.10 Game-Theoretic Shapley Attribution (Backup)

STAGE 1: Baseline
w=[0.5, 0.2, 0.2, 0.1]
Heuristic Prior

STAGE 2: Shapley
ϕ=[0.33, 0.28, 0.20, 0.
Exact Attribution

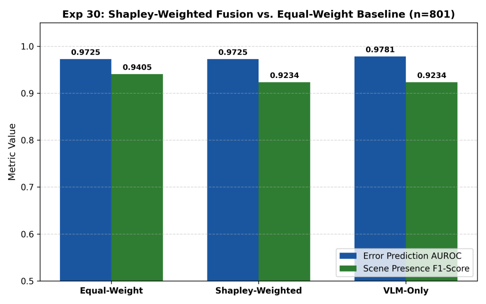
STAGE 3: Stacking
Platt Logistic
ECE $<$ **0.001**

Exp 30 Shapley Attribution Results

LLaVA-OneVision 7B ($\phi_1 = 0.330, 33.0\%$) and SAA Core ($\phi_2 = 0.283, 28.3\%$) provide over 61% of total diagnostic signal.

Not the Deployed Weighting

This is a supplementary interpretability study. Every result elsewhere in this thesis uses the fixed heuristic weights ($w = [0.6, 0.2, 0.2]$), not these Shapley values – Shapley attributions quantify contribution to a defined coalitional objective, not independent causal correctness.



Game-Theoretic Shapley Attribution Weights (ϕ_i).

- J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks," in *ICCV*, 2017.
- X. Wang, L. Xie, C. Dong, and Y. Shan, "Real-ESRGAN: Training Real-World Blind Super-Resolution with Pure Synthetic Data," in *ICCVW*, 2021.
- R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-Resolution Image Synthesis with Latent Diffusion Models," in *CVPR*, 2022.
- S. Liu et al., "Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection," in *ECCV*, 2023.
- B. Li et al., "LLaVA-OneVision: Easy Visual Task Transfer," *arXiv preprint arXiv:2408.03326*, 2024.

References ii

- L. S. Shapley, "A Value for n-Person Games," *Contributions to the Theory of Games*, 2(28):307–317, 1953.
- A. N. Angelopoulos and S. Bates, "Conformal Prediction: A Gentle Introduction," *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.